## The problem

Spain generates around 22 million tonnes of urban waste every year. According to Ecoembes, roughly one quarter to one third of what is placed in Spain's yellow recycling bin is sorted incorrectly, which forces manual cleaning of the stream at recycling plants and reduces the amount of packaging that can actually be recovered. The problem is not that families do not care. It is that the right information — which bin, which point, which rule — is hard to find at the exact moment of decision, standing in the kitchen with an empty container in hand.

In our own neighbourhood we ran a short survey of 31 residents of Majadahonda. 65% did not know where to take a used battery, and 70% could not name their nearest clean point. This matched our kitchen-table experience: even adults who want to recycle correctly often hesitate, and hesitation usually ends with the item in the wrong bin.

## The solution

Save the Planet by 3S is a mobile app that answers three questions in under thirty seconds: what to recycle, where to take it, and how much progress you are making. The app has five screens:

- Home — the user picks one of 24 territories in the Community of Madrid.
- Identify — the user types a waste item; the app normalises the text (lowercase, accent-folded) and uses a two-pass search (exact token match, then "contains" match) across 164 waste classes, returning container colour, collection-point type, and disposal instructions.
- Points — a filtered list and map of the nearest collection points, drawn from 3,186 special points and 34,824 common container locations organised into 218 chunks by district and waste type. One tap opens Google Maps with walking directions, using GPS only on button press and no API key.
- Impact — a weekly summary of sorted items and an early-stage behavioural $CO_2$ proxy (under active development; the complete per-category model is planned for the May 23 finals version).
- Report — the user can flag a missing or outdated point; flags are reviewed weekly against the source dataset.

The app is built in Thunkable with visual block programming and connected to a Google Sheets backend. All data comes from datos.comunidad.madrid (Community of Madrid open data portal) and is published under CC BY 4.0. To stay within Thunkable's data-loading limits we designed a chunk-manifest architecture: the 34,824 common points are split into 218 smaller tables by district and waste type, and the app queries only the one chunk needed for each search. This is the same kind of spatial indexing used by production GIS systems, adapted to run on a free no-code stack.

Our pilot with ten families over two weeks recorded 60% correct sorting before the app and 85% after, a 25-percentage-point observed improvement. We describe this honestly as a small, local pilot; the sample size (n=10) and duration (two weeks) are always shown alongside the result. Any broader adoption figure in our materials (for example, projected $CO_2$ reduction across Madrid) is clearly labelled as a scenario projection, not measured impact.

## Why it is a fit for the Climate Prize

The Climate Prize asks for a local climate-related problem, evidence of behaviour change, and support for informed decisions. Save the Planet by 3S addresses a local problem (recycling contamination in Madrid), shows a measured behaviour delta in a dated pilot with real families, and helps users make the right decision at the exact moment of action — which is the moment that actually changes outcomes downstream at the sorting plant.

## Honest limitations

The app is currently optimised for phones with a working data connection; an offline mode is planned for v2. The $CO_2$ impact screen uses a behavioural proxy that is under active development, and we say so in-app rather than presenting the number as a verified life-cycle assessment. Our pilot is promising but small; we will extend it after the May 23 finals version.

## Ethics statement

The app collects no personal data, creates no user accounts, and uses no tracking or analytics cookies. GPS is only accessed when the user presses "Open route"; it is never read in the background. All waste-classification data comes from public sources (datos.comunidad.madrid; Real Decreto 315/2025) and is re-published under CC BY 4.0 so other teams, councils, and researchers can reuse it without asking for permission. We designed data coverage to be

equal across all 24 Madrid territories, including lower-income districts, rather than optimising only for central areas. Where our $CO_2$ estimate is not yet fully verified we explicitly call it a behavioural proxy, in the app and in our submission texts, rather than presenting a preliminary number as a verified life-cycle assessment. Our mentors used AI tools (Claude and Open AI) to support Thunkable debugging and English-language review; we read, questioned, and validated every suggestion ourselves before using it.